

## Data bases in VB.Net

### Introduction

Today, people use computers to perform many tasks formerly done with other tools.

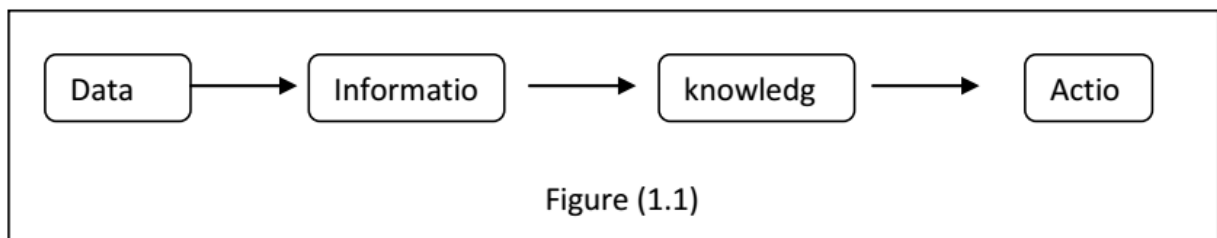
Computers have replaced typewriters for creating and modifying documents. They've also replaced millions of pieces of paper, file folders, and file cabinets as the principal storage medium for important information. Compared to those old tools, of course, computers do much more, much faster — and with greater accuracy. These increased benefits do come at a cost, however. Computer users no longer have direct physical access to their data. When computers occasionally fail, office workers may wonder whether computerization really improved anything at all. In the old days, a manila file folder only “crashed” if you dropped it then you merely knelt down, picked up the papers, and put them back in the folder. Barring earthquakes or other major disasters, file cabinets never “went down,” and they never gave you an error message.

### What is data?

- Data can be defined in many ways. Information science defines data as unprocessed information.

### What is information?

- Information is data that have been organized and communicated in a coherent and meaningful manner.
- Data is converted into information, and information is converted into knowledge.
- Knowledge is information evaluated and organized so that it can be used purposefully as shown in figure (1.1)



## What is a Data Base?

A **database** is an organized collection of data for one or more uses, typically in digital form. The data can be textual, like order or inventory data, or it can be pictures, programs or anything else that can be stored on a computer in binary form.

The purpose of a database is to store and retrieve related information, so databases are designed to offer an organized mechanism for:

- Storing.
- Managing.
- And retrieving information.

## Some basic Definitions:

**Field:** one category of information (one data value), i.e., Name, Address, Semester Grade, Academic topic.

**Record:** Collection of fields i.e., one student's information, a recipe, a test question.

**A File:** A group or collection of similar records, like student File.

**Digital databases** are managed using database management systems (DBMS), which store database contents, allowing data creation and maintenance, and search and other access to the database.

## What is DBMS?

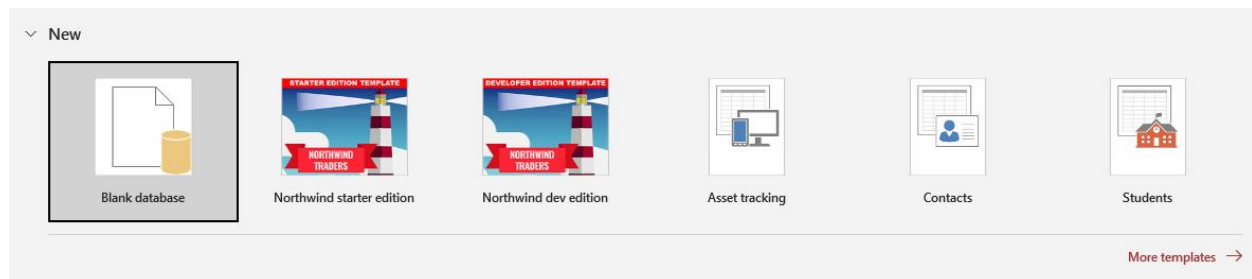
A Database Management System (DBMS) is a set of computer programs that controls the:

- Creation of the database.
- The storing and organization of the data in the database.
- Maintenance the database.
- Searching, data retrieval and the use of a database.

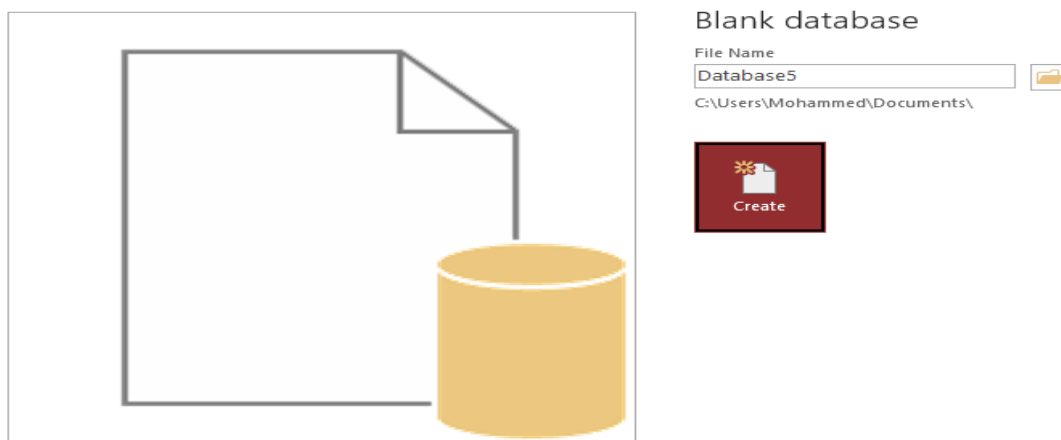
There is more than one type can connect with VB.Net like (Microsoft Access, Microsoft Excel, SQL and Oracle).

## How to Create Data base with Microsoft Access

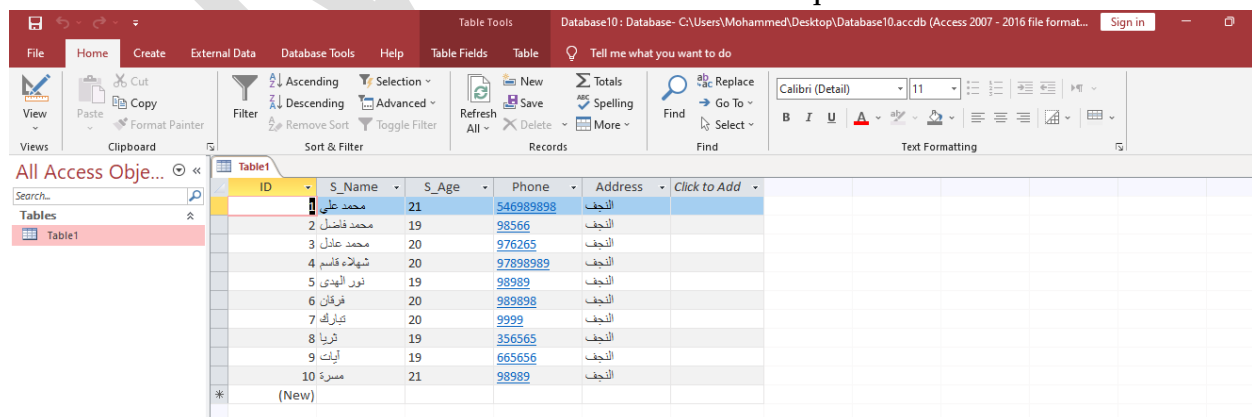
1. Enter to Microsoft Access in windows programs.
2. Select blank database.



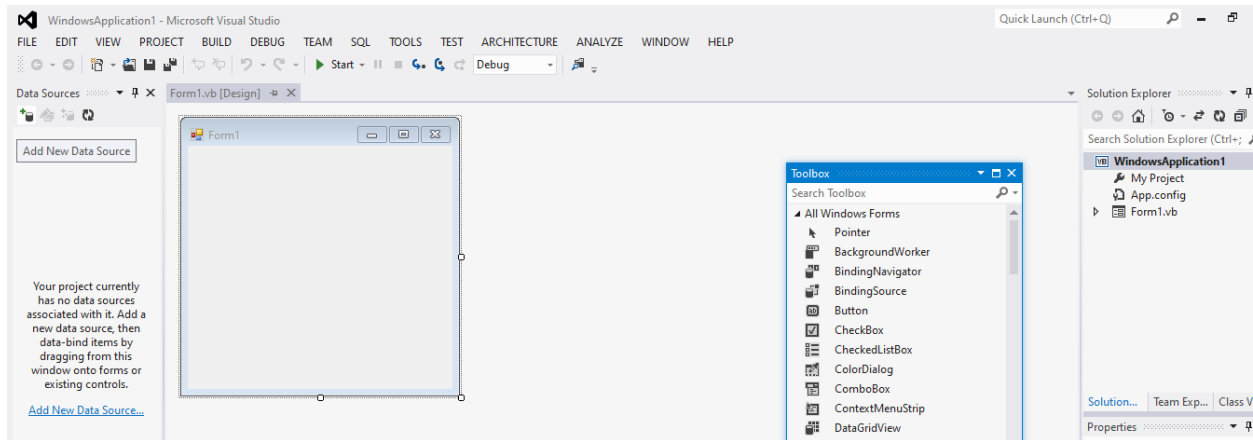
3. Enter file name for database and choose the location that save it, then click create.



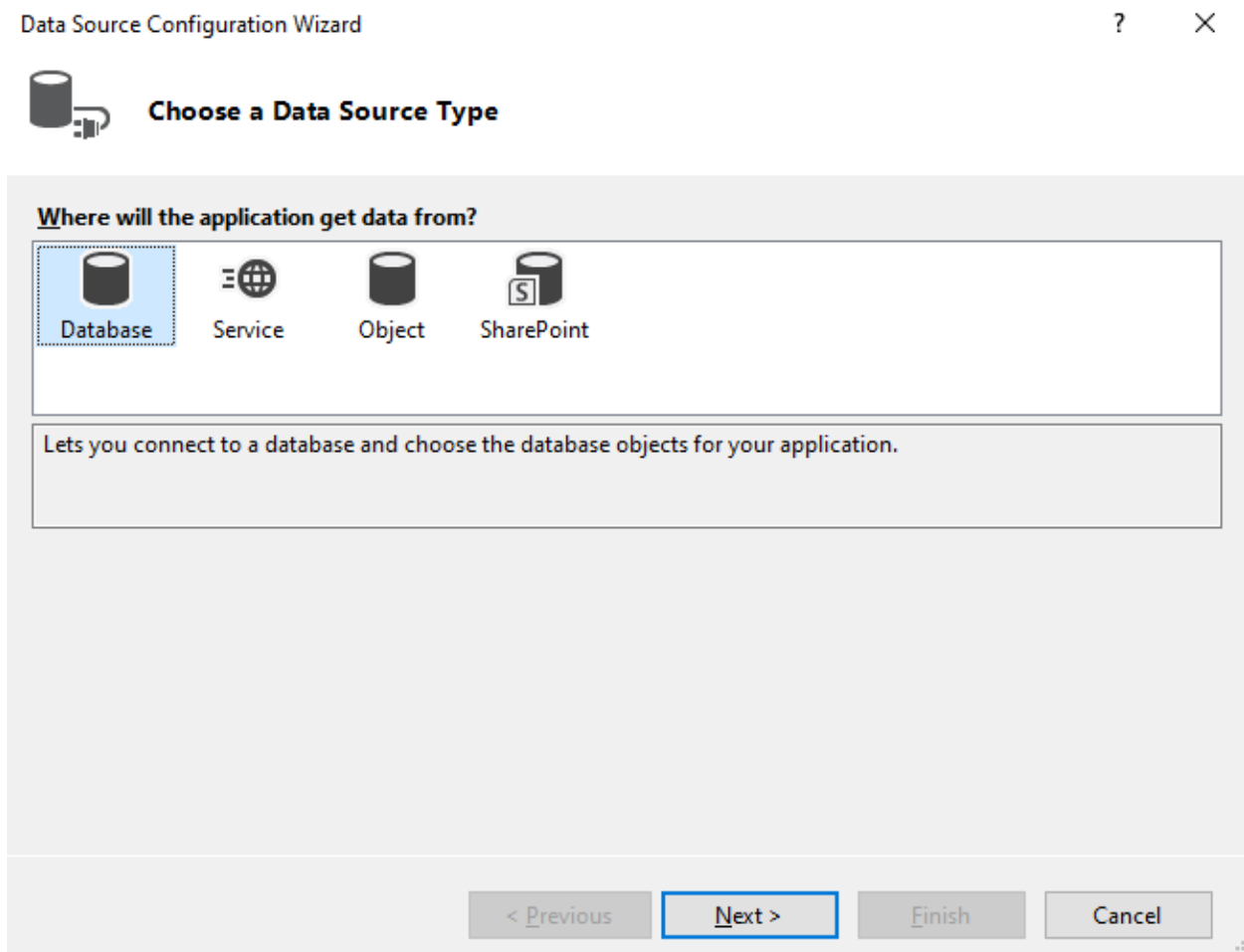
4. Enter the field name and data into data base that required.



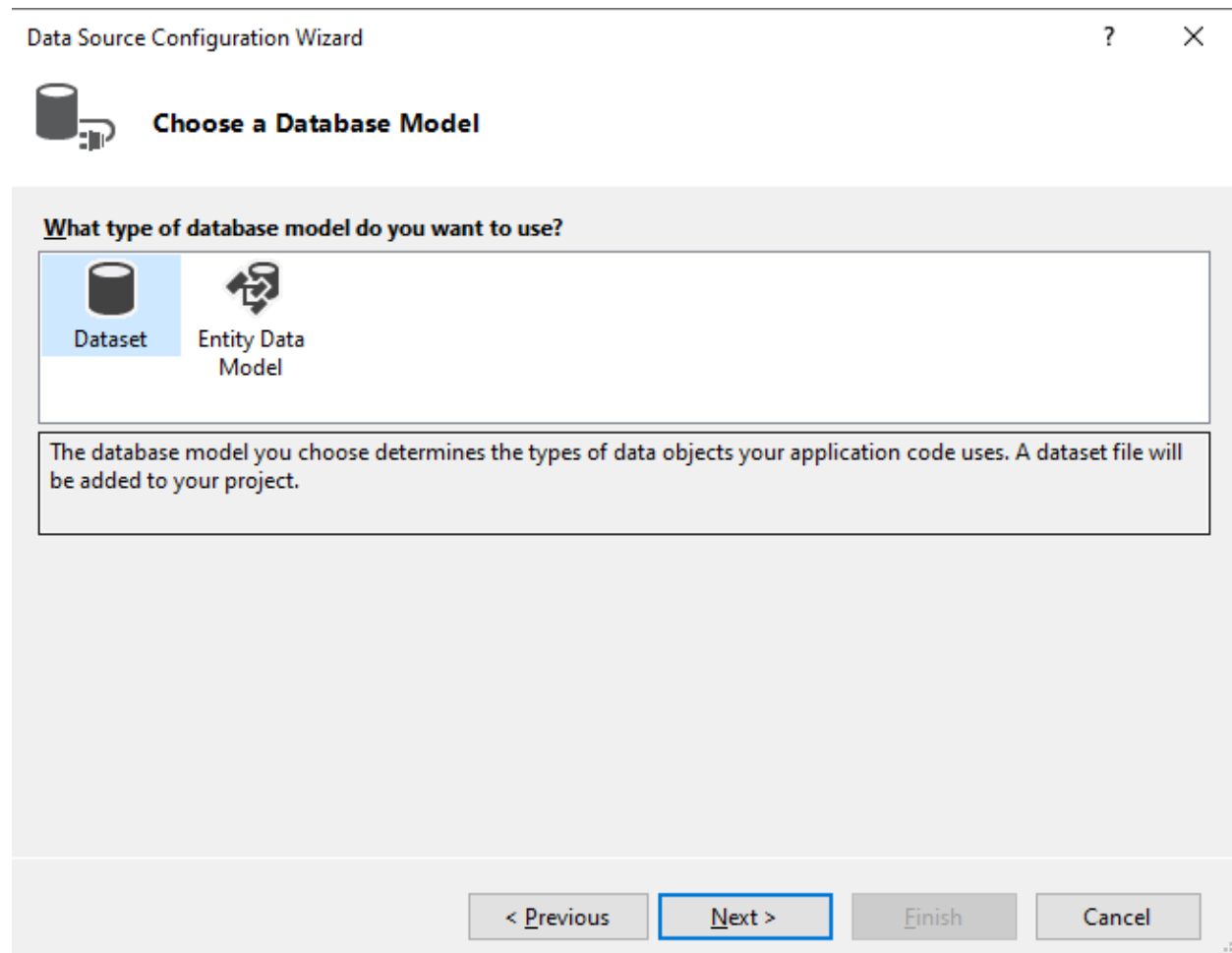
5. Save the file and exit Microsoft Access.
6. Create new project into VB.Net.
7. Add new data source from data source control.



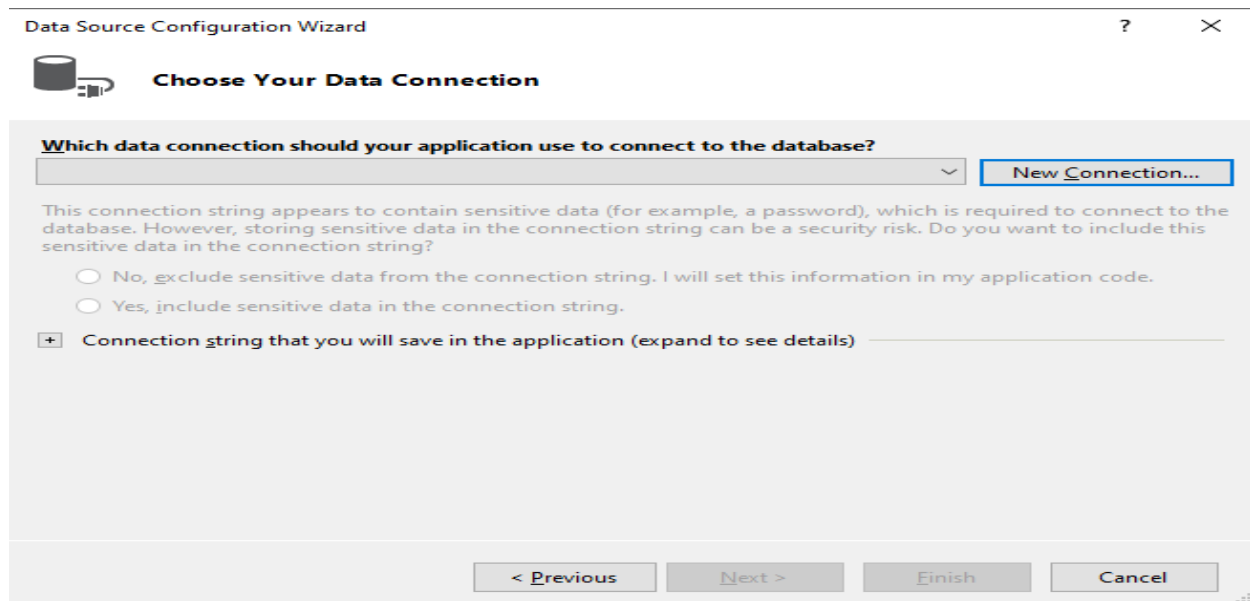
8. Choose a Data Source Type.



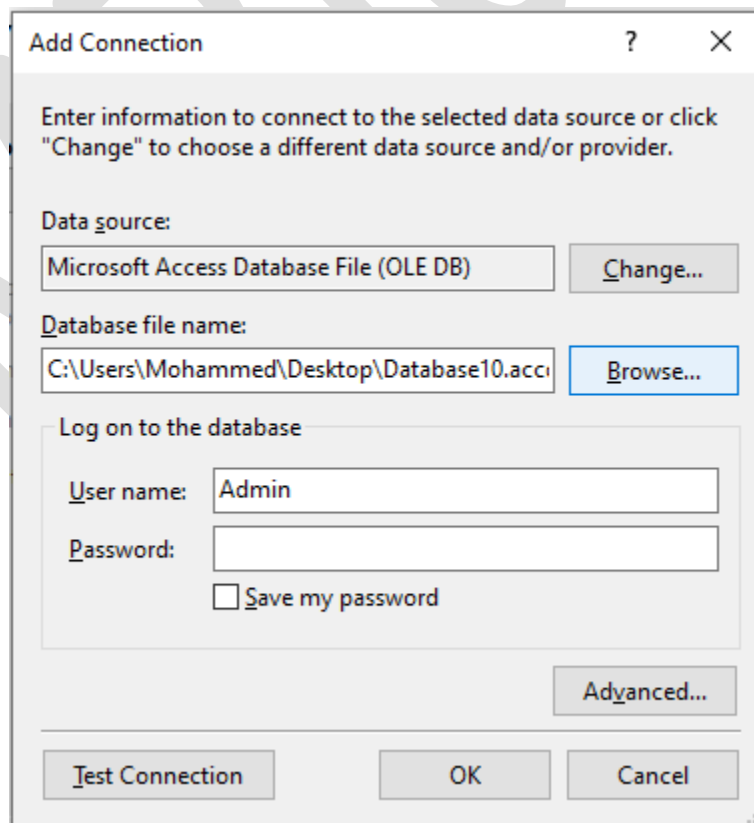
## 9. Choose a Database Model.



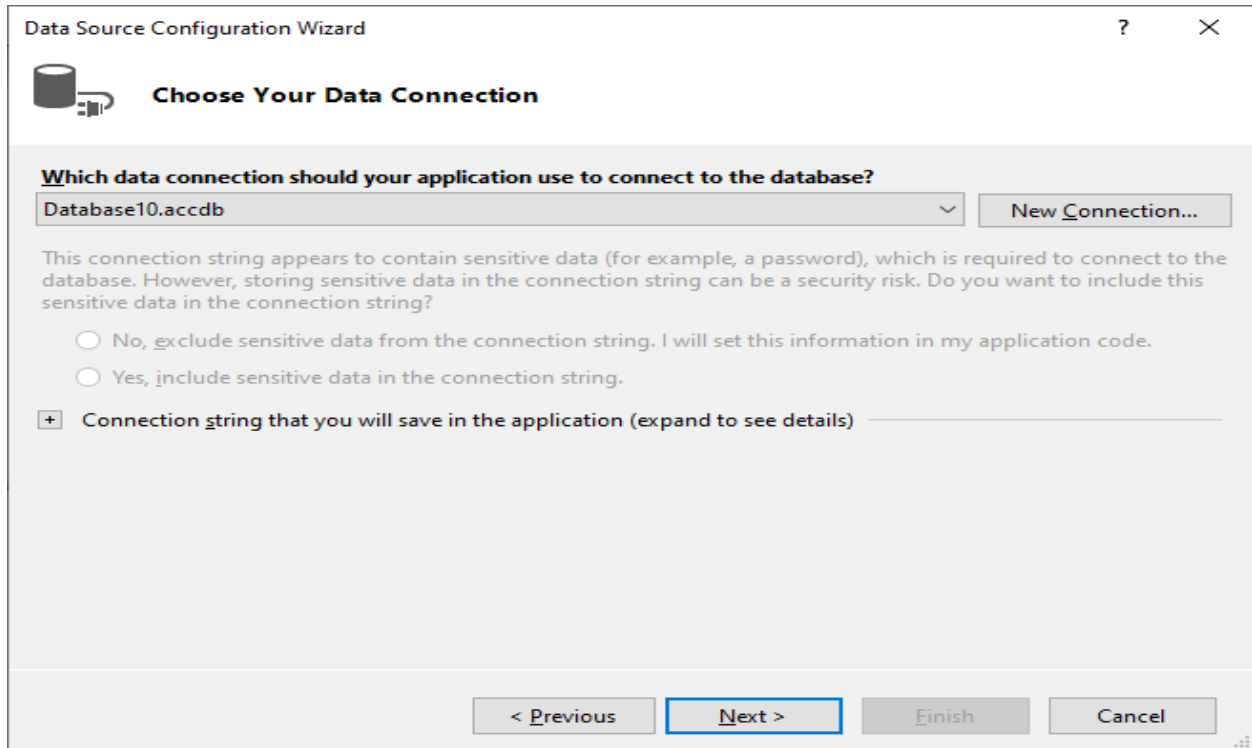
## 10. Choose Your Data Connection.



## 11. Click New Connection and click Browse to select Database File name, then click OK.

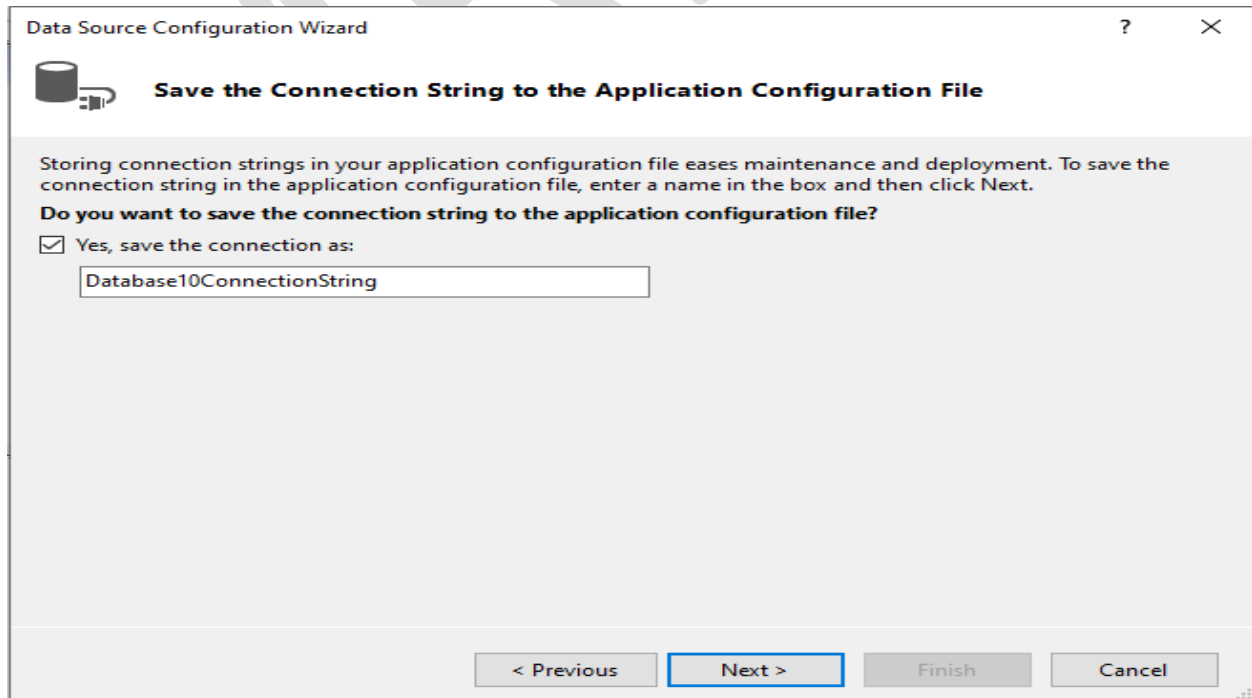


12. Choose Your Data Connection and click Yes.



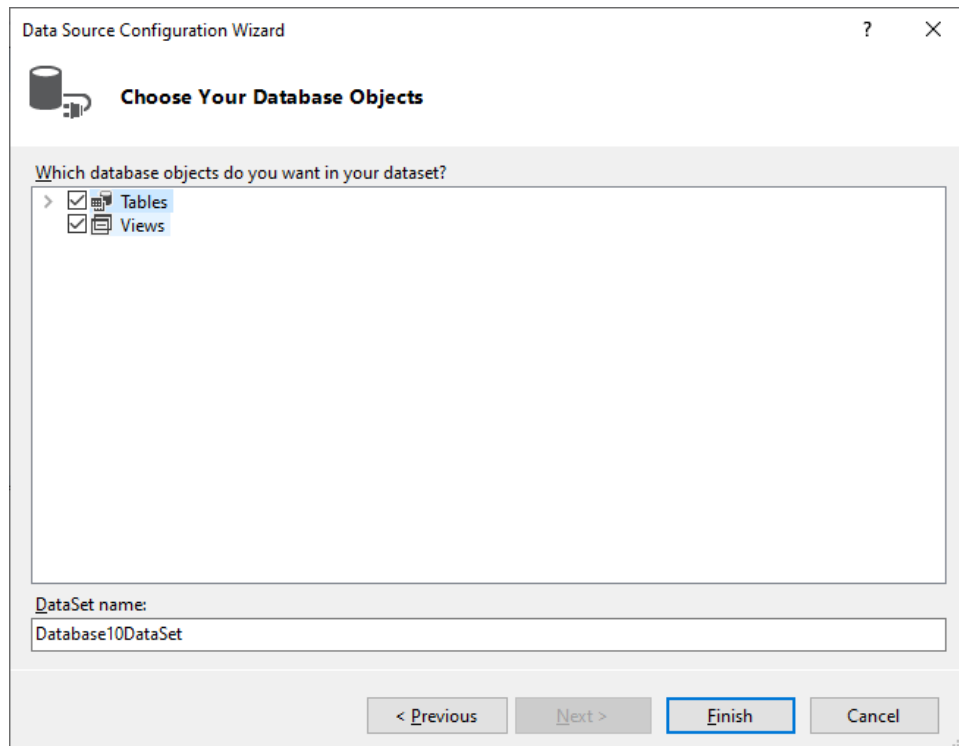
The screenshot shows the 'Data Source Configuration Wizard' window, specifically the 'Choose Your Data Connection' step. The window title is 'Data Source Configuration Wizard'. The main heading is 'Choose Your Data Connection'. Below the heading, there is a question: 'Which data connection should your application use to connect to the database?'. A dropdown menu shows 'Database10.accdb'. To the right of the dropdown is a button labeled 'New Connection...'. Below this, there is a paragraph of text: 'This connection string appears to contain sensitive data (for example, a password), which is required to connect to the database. However, storing sensitive data in the connection string can be a security risk. Do you want to include this sensitive data in the connection string?'. There are two radio buttons: 'No, exclude sensitive data from the connection string. I will set this information in my application code.' and 'Yes, include sensitive data in the connection string.'. Below the radio buttons is a plus sign icon and the text 'Connection string that you will save in the application (expand to see details)'. At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

13. Save the Connection.

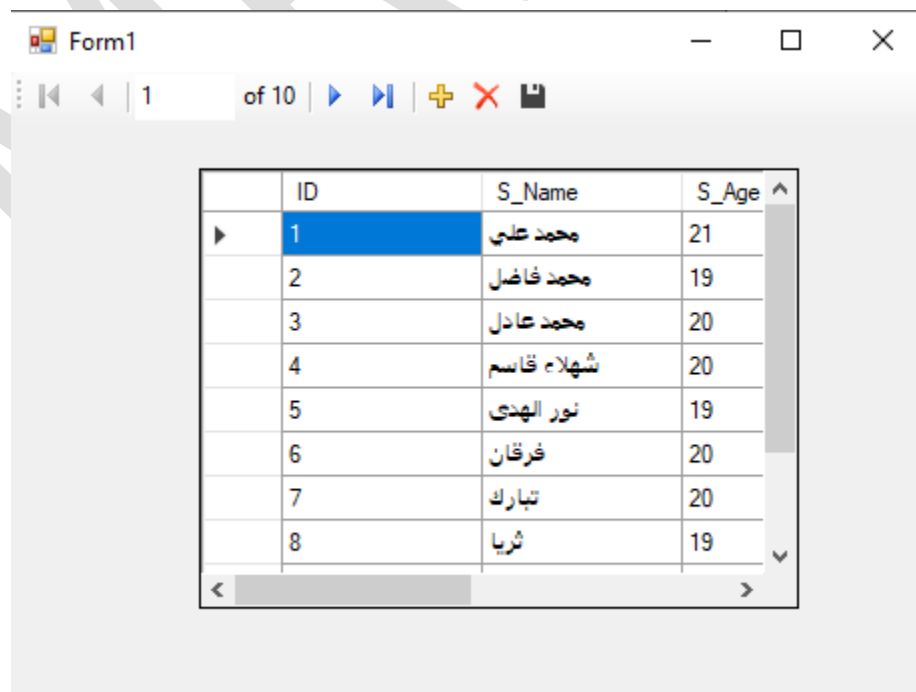


The screenshot shows the 'Data Source Configuration Wizard' window, specifically the 'Save the Connection String to the Application Configuration File' step. The window title is 'Data Source Configuration Wizard'. The main heading is 'Save the Connection String to the Application Configuration File'. Below the heading, there is a paragraph of text: 'Storing connection strings in your application configuration file eases maintenance and deployment. To save the connection string in the application configuration file, enter a name in the box and then click Next.' Below this, there is a question: 'Do you want to save the connection string to the application configuration file?'. There is a checked checkbox and the text 'Yes, save the connection as:'. Below this, there is a text box containing the text 'Database10ConnectionString'. At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

14. Choose Your Database Objects and click Finish.



15. Drag and Drop the Table1 on the Form, then run the project from start icon. Notes the data are display inside the grid.





16. Click the control to move Next, Previous, Move First and Move Last.
17. Click  to Add new Record.
18. Click  to Delete Current Record.
19. Click  to Save Record.